

DentalTensor

A Custom YOLO11n-Based AI Vision Model for Oral Pathology Screening

Research Paper Style Development Dossier

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Project Context: Conceived during the DaantShaant hackathon project and later productized as a reusable oral-image AI model integrated back into DaantShaant.

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Abstract

This document presents **DentalTensor**, a custom oral-image AI vision model built by Nathan Asif. DentalTensor was conceived while leading and building the **DaantShaant** hackathon product. Rather than relying only on a generic external model, a dedicated computer-vision pipeline was assembled, refined, calibrated, and integrated into the product. The work included multi-dataset auditing, hard-negative mining, manual review, v2 dataset construction, YOLO fine-tuning, calibration, deployment switching, and real UI validation on demonstration test images.

Highlight	Outcome
Model identity	DentalTensor — reusable custom oral-image AI vision model
Core architecture	Ultralytics YOLO11n (YOLO v11 nano family)
Base checkpoint	best.pt
Refined deployment checkpoint	best_v2.pt
Downloaded / audited source images	13,593 images across 3 datasets
Final v2 dataset size	10,806 images
Human-approved hard negatives	27 unique images
Effective negatives injected into v2 train	108
Deployment	Integrated into DaantShaant oral screening workflow

1. Project Background and Motivation

DaantShaant required a practical oral-screening component capable of analyzing user-uploaded mouth images and returning understandable findings. During project leadership and product development, the need for a more specialized and controllable computer-vision engine became clear. This led to the creation of DentalTensor as a reusable product/model of its own — a general-purpose oral-image analysis engine that could be integrated into DaantShaant and potentially reused elsewhere.

The guiding objective was not only to achieve model inference, but to engineer an end-to-end development workflow: source dataset auditing, false-positive analysis, negative mining, dataset curation, v2 training, calibration, deployment switching, and UI-level validation.

2. Dataset Sources and Counts

Dataset	Role	Downloaded / Audited Images	How It Was Used
oral-disease.yolov11	Primary labeled pathology dataset and base benchmark	10,698	Used as the full base dataset
Dental Data Set.yolov11	External Roboflow dataset mined for healthy-only controls / hard negatives	427	6 approved negatives selected
Penyakit Gigi Skripsi.yolov11	Large auxiliary healthy pool for false-positive mining and control isolation	2,468	13 approved negatives selected
TOTAL		13,593	Multi-dataset development pipeline

Primary dataset audit on **oral-disease.yolov11** established the baseline distribution: 10,698 total images, with 8,558 train, 1,070 valid, and 1,070 test images. This dataset contained 62,720 total boxes and showed a strong dominance of tooth-discoloration labels, which explained the model's tendency to over-trigger on healthy or mildly yellow teeth.

External datasets were not dumped blindly into training. Instead, they were mined carefully for safe controls and hard negatives. This is a strong point of the work: the auxiliary data was filtered through false-positive mining and human review rather than naïve merging.

3. Hard-Negative Mining and V2 Dataset Construction

A total of **27** unique hard-negative / control images were approved after review across the curated CSV pipelines. These came from three review stages:

- 8 approved hard-negative images from review_ai_recommended.csv
- 6 approved hard-negative / clean-control images from review_healthy_ai_recommended.csv
- 13 approved hard-negative / clean-control images from review_final_large_pool_ai_curated.csv

To strengthen their impact during training, the negatives were oversampled at **4×**, creating **108** effective negative training instances in the v2 train split. Leakage protection was enforced by hashing untouched validation and test splits before the merge.

Split	Image Count	Notes
Train	8,666	Includes 108 injected hard-negative instances
Valid	1,070	Kept 100% untouched for benchmark integrity
Test	1,070	Kept 100% untouched for benchmark integrity
Total	10,806	Negative fraction in train ≈ 1.25%

4. Model Training, Fine-Tuning, and Deployment

DentalTensor was built around **Ultralytics YOLO11n (YOLO v11 nano family)**. The base oral-pathology checkpoint **best.pt** was refined into **best_v2.pt** using the prepared v2 dataset. Training was run in the cloud using a Modal runtime with an **NVIDIA A10 GPU**.

Model summary captured during the process: **101 layers, 2,583,127 parameters, 6.4 GFLOPs**. The final trained weight file was downloaded and placed at **services/teeth_analyzer/models/oral_disease/best_v2.pt**, then wired into the runtime so the application would load the refined model instead of the earlier checkpoint.

This matters because the work was not only academic experimentation. It was an end-to-end product integration exercise: train, export, swap the active model path, preserve deployment compatibility, and validate behavior inside the real user interface.

5. Calibration Strategy

A calibration report was generated on the untouched validation split. Engineering analysis indicated that best-F1 operating points tended to sit near 0.30, but the deployed product configuration used more conservative class-specific thresholds to reduce false positives in user-facing operation.

Detection Threshold	Configured Value
Global fallback	0.50
Calculus	0.35
Caries	0.55
Gingivitis	0.50
Tooth discoloration	0.65
Ulcer	0.65

Final deployment therefore combined two improvements: (1) a refined v2 model checkpoint and (2) better threshold policy tuned per class. This is what made the UI screenshots meaningful evidence rather than just raw model demos.

6. Qualitative Demonstration Results

The four user-facing test cases below demonstrate the deployed DentalTensor experience inside the DaantShaant interface. These screenshots are intentionally shown as product evidence: the model was not only trained, it was operationalized with confidence scores, summaries, recommended action, and next-step guidance.

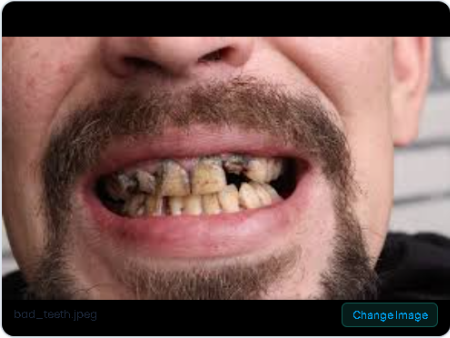
Case 1 — Severe / visibly compromised dentition

Oral Scan

Snapshot

Live

Upload



bad_teeth.jpeg

Change Image

Clear

Analyze Upload

Analysis complete

Oral Health Screening Report

SCREENING SUMMARY

Possible early tooth decay concern

85%

URGENCY: DENTAL VISIT RECOMMENDED

RECOMMENDED ACTION

Arrange a licensed dental examination to confirm or rule out decay

RECOMMENDED TIMEFRAME

Arrange a dental evaluation soon

RECOMMENDED SPECIALIST

general dentist

SCREENING OVERVIEW

AI screening suggests arranging a dental evaluation soon

POSSIBLE CONCERNS

- Visual feature possibly consistent with early tooth decay

SCREENING LIMITATIONS

- Screening uses only what is visible in a single photo; it cannot detect pain, swelling, or conditions that need an examination or x-rays.
- Moderate screening confidence (below internal review threshold) — professional confirmation is recommended.

Observed result: Possible early tooth decay concern **Confidence:** 85%

The deployed DentalTensor-powered flow correctly escalated the visibly abnormal image for a dental evaluation soon.

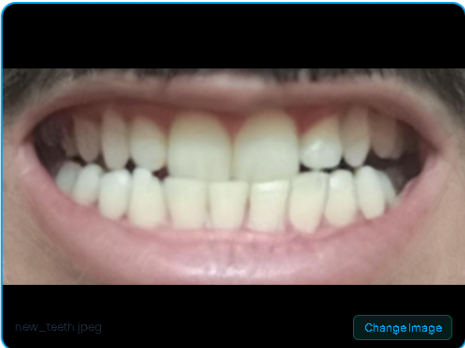
Case 2 — Normal-looking smile (new_teeth)

Oral Scan

Snapshot

Live

Upload



new_teeth.jpeg

Change Image

Clear

Analyze Upload

Analysis complete

Oral Health Screening Report

SCREENING SUMMARY

No obvious concerning visual finding

85%

URGENCY: ROUTINE

RECOMMENDED ACTION

Maintain good oral hygiene (brush twice daily, clean between teeth)

RECOMMENDED TIMEFRAME

Routine dental checkup

RECOMMENDED SPECIALIST

general dentist

SCREENING OVERVIEW

AI screening found no obvious concerning visual finding - routine monitoring

POSSIBLE CONCERNS

- No obvious concerning visual finding

RECOMMENDED NEXT STEPS

- Continue routine dental checkups

SCREENING LIMITATIONS

- Screening uses only what is visible in a single photo; it cannot detect pain, swelling, or conditions that need an examination or x-rays.

Observed result: No obvious concerning visual finding **Confidence:** 85%

Healthy-looking dentition was correctly kept in the routine-monitoring category.

Case 3 — Yellowish teeth / discoloration example

Oral Scan

Snapshot Live **Upload**



yellowish-teeth.jpg **Change Image**

Clear

Analyze Upload

Analysis complete

Oral Health Screening Report

SCREENING SUMMARY



Possible tooth discoloration

76%

URGENCY: ROUTINE

RECOMMENDED ACTION

Maintain oral hygiene and monitor the discoloration

RECOMMENDED TIMEFRAME

Routine dental checkup

RECOMMENDED SPECIALIST

general dentist

SCREENING OVERVIEW

AI screening suggests routine monitoring

POSSIBLE CONCERNS

- Possible tooth discoloration

RECOMMENDED NEXT STEPS

- Arrange a dental evaluation if persistent, changing, or concerning

SCREENING LIMITATIONS

- Screening uses only what is visible in a single photo; it cannot detect pain, swelling, or conditions that need an examination or x-rays.
- This visual screening cannot determine the underlying cause of tooth

Observed result: Possible tooth discoloration **Confidence:** 76%
DentalTensor correctly separated discoloration-like appearance from the healthy-control cases.

Case 4 — Good / bright teeth control

Oral Scan

Snapshot Live **Upload**



good-teeth.jpg **Change Image**

Clear

Analyze Upload

Analysis complete

Oral Health Screening Report

SCREENING SUMMARY



No obvious concerning visual finding

85%

URGENCY: ROUTINE

RECOMMENDED ACTION

Maintain good oral hygiene (brush twice daily, clean between teeth)

RECOMMENDED TIMEFRAME

Routine dental checkup

RECOMMENDED SPECIALIST

general dentist

SCREENING OVERVIEW

AI screening found no obvious concerning visual finding - routine monitoring

POSSIBLE CONCERNS

- No obvious concerning visual finding

RECOMMENDED NEXT STEPS

- Continue routine dental checkups

SCREENING LIMITATIONS

- Screening uses only what is visible in a single photo; it cannot detect pain,

Observed result: No obvious concerning visual finding **Confidence:** 85%
A clean control case stayed negative after the v2 model swap and threshold tuning.

7. Why DentalTensor Is a Strong Project

- Built a custom vision model pipeline rather than stopping at prompt-level AI usage.
- Audited a large primary YOLO-format dataset and identified class imbalance and false-positive patterns.
- Extended the base dataset with human-reviewed hard negatives instead of doing a naïve retrain.
- Protected benchmark validity by keeping validation and test splits untouched and checking leakage.
- Ran cloud fine-tuning, exported the trained weights, and integrated the refined model into a live product.
- Calibrated the deployment with class-specific thresholds and verified behavior through user-facing cases.
- Turned a hackathon need into a reusable product concept: DentalTensor, then implemented it inside DaantShaant.

8. Product Positioning

DentalTensor can be presented as a standalone AI vision asset developed by Nathan Asif: a custom oral-screening model capable of detecting dental pathology patterns from mouth photographs and powering screening experiences in healthcare or health-tech products. DaantShaant is the first product implementation and validation environment for the model.

9. References / Development Sources

- Primary dataset: oral-disease.yolov11 (local YOLO-format oral pathology dataset used as the base benchmark and training source).
- Auxiliary dataset: Dental Data Set.yolov11 (427-image Roboflow export; healthy-only controls mined under review).
- Auxiliary dataset: Penyakit Gigi Skripsi.yolov11 (2,468-image Roboflow export; source project penyakit-gigi-skripsi-i77mi; CC BY 4.0; used for large healthy-pool mining and control isolation).
- Model family: Ultralytics YOLO11n.
- Training / refinement scripts discussed in project workflow: dataset audit, hard-negative mining, v2 dataset preparation, Modal training, and calibration evaluation.

10. Closing Statement

In summary, Nathan Asif did more than build a prototype interface. He built a custom AI model pipeline, curated data from multiple sources, refined the detector with reviewed hard negatives, trained and deployed a better checkpoint, tuned thresholds, and proved the result inside a working application. DentalTensor is therefore both a research-style technical achievement and a product-level engineering accomplishment.

Note: DentalTensor is presented here as a screening-support system for demonstration and research/development purposes. It is not a substitute for diagnosis by a licensed dental professional.